5.3.6 Differentiation of a function with respect to another function

Let $u = f(x)$ and $v = g(x)$ be two functions of x. The derivative of $f(x)$ with respect to $g(x)$ is given by the formula,

$$\frac{d(f(x))}{d(g(x))} = \frac{du/dx}{dv/dx}$$

Example 5.43

Find $\frac{dy}{dx}$ if

(i) $x = at^2, y = 2at$

(ii) $x = a\cos\theta,\ y = a\sin\theta$

Solution

(i) $x = at^2 \quad | \quad y = 2at$

$\frac{dx}{dt} = 2at \quad | \quad \frac{dy}{dt} = 2a$

$$\therefore \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$= \frac{2a}{2at}$$

$$= \frac{1}{t}$$

(ii) $x = a\cos\theta,\ y = a\sin\theta$

$\frac{dx}{d\theta} = -a\sin\theta \quad | \quad \frac{dy}{d\theta} = a\cos\theta$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{a\cos\theta}{-a\sin\theta}$$

$$= -\cot\theta$$

Example 5.44

If $x = a\theta$ and $y = \frac{a}{\theta}$, then prove that $\frac{dy}{dx} + \frac{y}{x} = 0$

Solution

$x = a\theta \quad | \quad y = \frac{a}{\theta}$

$\frac{dx}{d\theta} = a \quad | \quad \frac{dy}{d\theta} = \frac{-a}{\theta^2}$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{\left(\frac{-a}{\theta^2}\right)}{a}$$

$$= -\frac{1}{\theta^2}$$

$$= -\frac{y}{x}$$

i.e. $\frac{dy}{dx} + \frac{y}{x} = 0$

Aliter :

Take $xy = a\theta \cdot \frac{a}{\theta}$

$$xy = a^2$$

Differentiating with respect to x,

$$x\frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} + \frac{y}{x} = 0$$

Example 5.45

Differentiate $\frac{x^2}{1+x^2}$ with respect to x^2

Solution

Let $u = \frac{x^2}{1+x^2}$ and $v = x^2$

$\frac{du}{dx} = \frac{(1+x^2)(2x) - x^2(2x)}{(1+x^2)^2} \quad | \quad \therefore \frac{dv}{dx} = 2x$

$$= \frac{2x}{(1+x^2)^2}$$

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